



SYDNEY GRAMMAR SCHOOL
MATHEMATICS DEPARTMENT
YEARLY EXAMINATIONS 2007

FORM V

MATHEMATICS

+ Solutions

Y11 24 Yearly
2007

Syd Gram 2007 Y11 24 Q&S

Examination date

Thursday 18th October 2007

Time allowed

2 hours

Instructions

- All nine questions may be attempted.
- All nine questions are of equal value.
- All necessary working must be shown.
- Marks may not be awarded for careless or badly arranged work.
- Start each question on a new booklet.
- Approved calculators and templates may be used.

Collection

- Write your name, class and master clearly on each booklet.
- Hand in the nine questions in a single well-ordered pile.
- Hand in a booklet for each question, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Keep the printed examination paper and bring it to your next Mathematics lesson.

5P: TCW 5Q: RCF 5R: DNW

Checklist

- SGS booklets: 9 per boy. A total of 500 booklets should be sufficient.
- Candidature: 45 boys.

Examiner

TCW

QUESTION ONE (12 marks) Use a separate writing booklet.

(a) Simplify:

(i) $(2\sqrt{3})^2$

(ii) $8^{-\frac{1}{3}}$

(b) Solve:

(i) $|x| = 3$

(ii) $x(x + 10) = 0$

(c) Evaluate:

(i) $\log_6 1$

(ii) $\log_6 180 - \log_6 5$

(d) (i) Find the exact value of $\sin 225^\circ$.

(ii) Find the acute α , correct to the nearest minute, given that $\cot \alpha = \frac{1}{3}$.

(e) Factorise:

(i) $x^2 - 100$

(ii) $x^3 - 1000$

QUESTION TWO (12 marks) Use a separate writing booklet.

(a) Rationalise the denominator then simplify: $\frac{10}{\sqrt{5} - \sqrt{3}}$.

(b) (i) Write $y = x^2 + 4x + 7$ in the form $y = (x - h)^2 + k$ by completing the square.

(ii) Hence or otherwise write down the coordinates of the vertex of the parabola $y = x^2 + 4x + 7$.

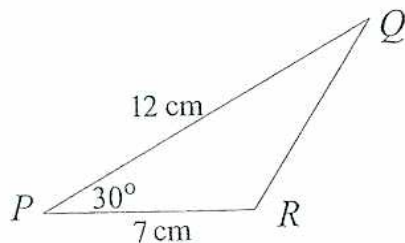
(c) Differentiate:

(i) $y = 5x^4$

(ii) $y = 2\sqrt{x}$

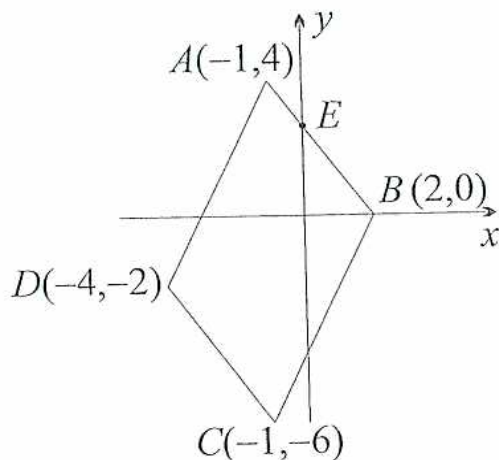
(iii) $y = \frac{6x^2 - 14x^5}{2x}$

(d)



Find the area of $\triangle PQR$ in the diagram above.

QUESTION THREE (12 marks) Use a separate writing booklet.



The diagram above shows the parallelogram $ABCD$ with vertices $A(-1, 4)$, $B(2, 0)$, $C(-1, -6)$ and $D(-4, -2)$. The point E is the y -intercept of the line AB .

- Show that $AB \parallel DC$.
- Show that the equation of the line AB is $4x + 3y - 8 = 0$.
- Find the coordinates of E .
- Find the coordinates of the midpoint M of AB .
- Find the equation of the circle with centre A and radius AB .
- Show that the perpendicular distance from D to AB is 6 units.
- Hence find the area of parallelogram $ABCD$.
Note: The area of a parallelogram is given by:
Area = Base $\times$ Perpendicular Height.

QUESTION FOUR (12 marks) Use a separate writing booklet.

- (a) Use the quadratic formula to solve $2x^2 - 6x - 3 = 0$.
- (b) For what values of c is the quadratic $2x^2 - 3x + c$ positive definite?
- (c) For what values of m does the equation $2x^2 - mx + 2 = 0$ have two distinct real roots?
- (d) If α and β are the zeroes of $-x^2 - 4x + 10$, find:
 - (i) $\alpha + \beta$
 - (ii) $\alpha\beta$
 - (iii) $(\alpha + 2)(\beta + 2)$
 - (iv) $\alpha^2 + \alpha\beta + \beta^2$

QUESTION FIVE (12 marks) Use a separate writing booklet.

- (a) A point $P(x, y)$ moves so that it is always 3 units from the line $x = 1$. Find the equations of the locus of P .
- (b) Consider the parabola with equation $x^2 = 12y$.
 - (i) Write down the coordinates of the vertex.
 - (ii) Write down the coordinates of the focus.
 - (iii) Write down the equation of the directrix.
- (c) An arithmetic sequence has first term 15 and common difference 25.
 - (i) Write out the first three terms of the sequence.
 - (ii) Find the sum of the first 41 terms.
- (d) In a certain athletics season, each time Rory runs an 800 metre race he lowers his time by $2\frac{1}{4}$ seconds. In his first 800 metre race of the season his time is 2 minutes 30 seconds.
 - (i) Rory's successive race times in seconds form an arithmetic sequence. Show that the n th term in the sequence is $T_n = \frac{609 - 9n}{4}$.
 - (ii) How many races will he have to run throughout the season to lower his time below the magic 1 minute 50 second barrier?

QUESTION SIX (12 marks) Use a separate writing booklet.

- (a) Sketch the graph of $y = \log_5 x$, clearly showing the asymptote, the x -intercept and one other point.
- (b) Find the equation of the ~~parabola with~~ vertex $(-2, 1)$ and focus $(-2, 3)$.
- (c) (i) Differentiate $y = (x + 3)^{10}$.
- (ii) Use the quotient rule to find the derivative of $y = \frac{x^2}{x - 2}$.
- (iii) (α) Use the product rule to show that $\frac{d}{dx} (5x(x - 4)^4) = 5(5x - 4)(x - 4)^3$.
- (β) Hence find the x -coordinates of the points on the curve $y = 5x(x - 4)^4$ where the tangents to the curve are horizontal.

QUESTION SEVEN (12 marks) Use a separate writing booklet.

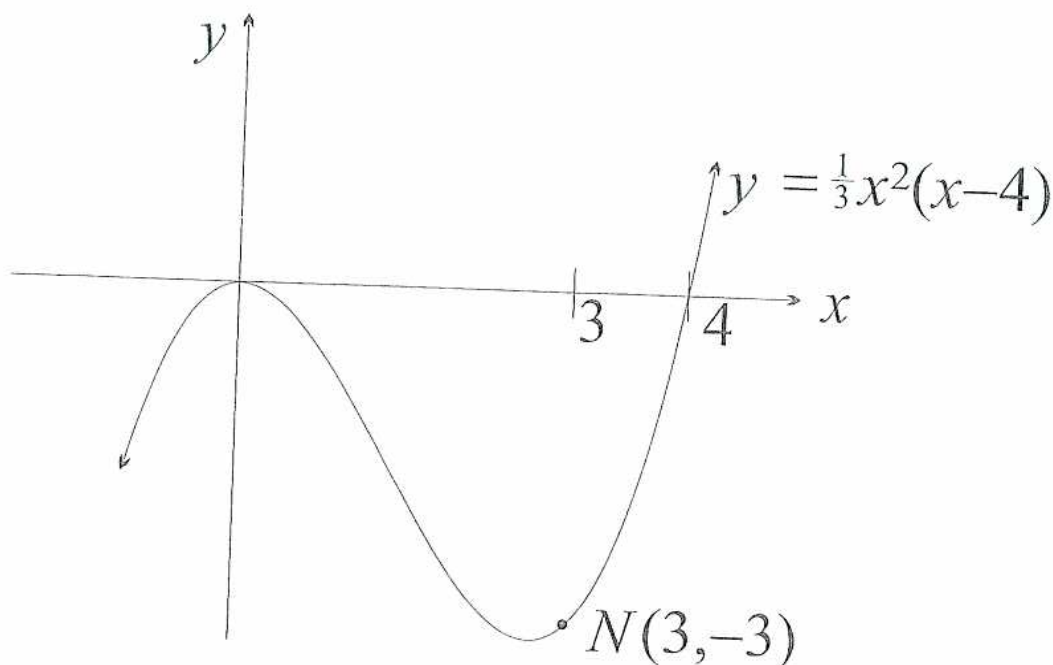
- (a) In $\triangle TUV$, $\angle T = 120^\circ$, $TU = 10$ cm and $TV = 15$ cm.
- (i) Draw a diagram representing $\triangle TUV$, clearly labelling the vertices and marking the given side lengths and $\angle T$.
- (ii) Show that the exact side length of UV is $5\sqrt{19}$ cm.
- (b) Evaluate: $\sum_{n=1}^{20} 3 \times 2^{n-1}$.
- (c) The fourth term in a geometric sequence is 4 and the eleventh term is -512 . Find the eighth term.
- (d) Without finding the point of intersection, find the equation of the line through the point $(1, -3)$ that also passes through the intersection of the two lines $4x - 3y + 2 = 0$ and $x - 2y - 3 = 0$. Give your answer in simplest general form.

QUESTION EIGHT (12 marks) Use a separate writing booklet.

(a) Solve $\cos \theta = 1$ for $0^\circ \leq \theta \leq 360^\circ$.

(b) Solve $2 \log_{10} x = \log_{10}(6 - x)$.

(c)



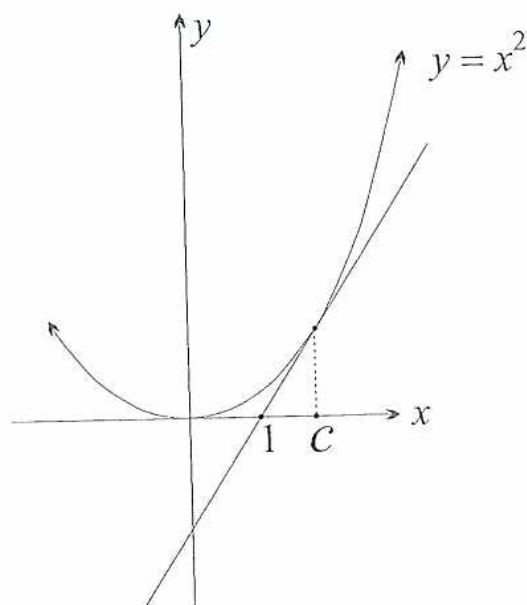
The diagram above shows the graph of $y = \frac{1}{3}x^2(x - 4)$. N is a point on the curve with coordinates $(3, -3)$.

- (i) Show that the tangent to $y = \frac{1}{3}x^2(x - 4)$ at N has equation $y = x - 6$.
- (ii) Show that the normal to $y = \frac{1}{3}x^2(x - 4)$ at N has equation $x + y = 0$.
- (iii) The points W and X are the points where the tangent and normal at N meet the x -axis. Find the area of $\triangle WXN$.

QUESTION NINE (12 marks) Use a separate writing booklet.

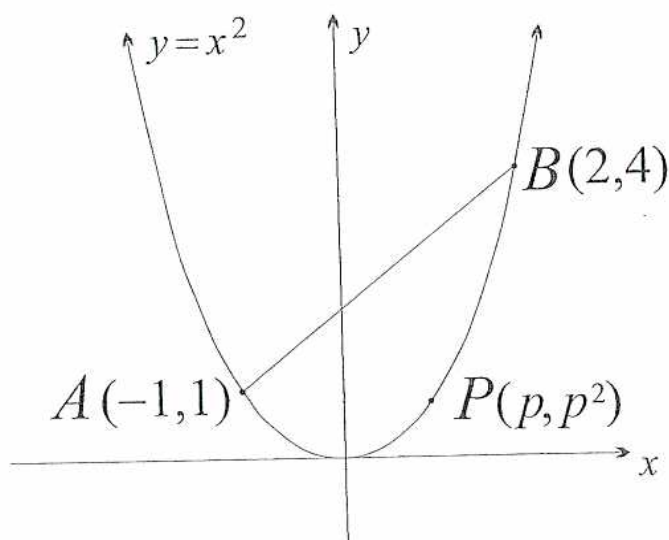
- (a) Find the first three terms of the geometric sequence that has a common ratio of $\frac{1}{t}$ and a limiting sum of $\frac{1}{1-t}$.

(b)



The diagram above shows a graph of the parabola $y = x^2$, and the tangent to the parabola at the point where $x = c$, for $c \neq 0$. For what value of c does the tangent intersect the x -axis at $x = 1$? Clearly show your working.

(c)



In the diagram above $A(-1, 1)$ and $B(2, 4)$ are points on the parabola $y = x^2$. The point $P(p, p^2)$ is a variable point on the parabola that lies below the line AB .

- (i) Show that line AB has equation $x - y + 2 = 0$.
- (ii) Find the maximum possible area of $\triangle APB$.

END OF EXAMINATION

Q1

(a) (i) $(2\sqrt{3})^2 = 12$ ✓

(ii) $8^{-\frac{1}{3}} = \frac{1}{2}$ ✓

(b) (i) $|x| = 3$ ✓

$x = 3$ or -3 ✓

(ii) $x(x+10) = 0$ ✓

$x = 0$ or -10 ✓

(c) (i) $\log_6 1 = 0$ ✓

(ii) $\log_6 180 - \log_6 5 = \log_6 \frac{180}{5}$ ✓
 $= \log_6 36$ ✓
 $= 2$ ✓

(d) (i) $\sin 225^\circ = -\sin 45^\circ$ ✓
 $= -\frac{1}{\sqrt{2}}$ ✓

(ii) $\cot \alpha = \frac{1}{3}$ ✓

$\tan \alpha = 3$ ✓

[no penalty for rounding] $\alpha \doteq 71^\circ 34'$ ✓
 (nearest minute)

(e) (i) $x^2 - 100 = (x-10)(x+10)$ ✓

(ii) $x^3 - 1000 = x^3 - 10^3$ ✓
 $= (x-10)(x^2+10x+100)$ ✓

Q2

(a) $\frac{10}{\sqrt{5}-\sqrt{3}} = \frac{10}{\sqrt{5}-\sqrt{3}} \times \frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}+\sqrt{3}}$ ✓
 $= \frac{10(\sqrt{5}+\sqrt{3})}{2}$

$= 5(\sqrt{5}+\sqrt{3})$ ✓

[or $5\sqrt{5} + 5\sqrt{3}$]

(b) (i) $y = x^2 + 4x + 7$ ✓

$y = x^2 + 4x + 4 + 3$ ✓

$y = (x+2)^2 + 3$ ✓✓

(ii) vertex = $(-2, 3)$ ✓

(c) (i) $y = 5x^4$ ✓

$y' = 20x^3$ ✓

(ii) $y = 2x^{\frac{1}{2}}$ ✓

$y' = 2 \times \frac{1}{2} x^{-\frac{1}{2}}$ ✓

$= x^{-\frac{1}{2}}$ ✓
 $= \frac{1}{\sqrt{x}}$

(iii) $y = \frac{6x^2}{2x} - \frac{14x^5}{2x}$ ✓

$y = 3x - 7x^4$ ✓

$y' = 3 - 28x^3$ ✓

(d) Area = $\frac{1}{2} \times 7 \times 12 \times \sin 30^\circ$ ✓
 $= 21 \text{ cm}^2$ ✓

[-1 for incorrect or no units]

Q3 A(-1,4) B(2,0) C(-1,-6) D(-4,-2)

(a) $m_{AB} = \frac{0-4}{2+1} = -\frac{4}{3}$ $m_{DC} = \frac{-2+6}{-4+1} = -\frac{4}{3}$ ✓

$m_{AB} = m_{DC}$, so $AB \parallel DC$.

(b) AB: $y-0 = -\frac{4}{3}(x-2)$ ✓
 $3y = -4x + 8$ ✓
 $4x + 3y - 8 = 0$

(c) when $x=0$, $3y=8$ ✓
 $y = \frac{8}{3}$ ✓
 $E = (0, 2\frac{2}{3})$ ✓

(d) $M_{AB} = (\frac{2-1}{2}, \frac{0+4}{2})$ ✓
 $= (\frac{1}{2}, 2)$ ✓

(e) centre A(-1,4) ✓
 $AB^2 = 3^2 + 4^2$ ✓
 $AB = 5$ ✓
 circle: $(x+1)^2 + (y-4)^2 = 25$ ✓

(f) Perp. distance D to AB ✓
 $= \frac{|4(-4) + 3(-2) - 8|}{\sqrt{4^2 + 3^2}}$ ✓
 $= \frac{|-30|}{5}$ ✓
 $= 6$ units ✓

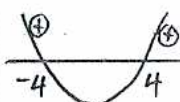
(g) Area = $6 \times AB$ ✓
 $= 6 \times 5$ ✓
 $= 30$ units² ✓

Q4

(a) $2x^2 - 6x - 3 = 0$ ✓
 $x = \frac{6 \pm \sqrt{36 - 4(2)(-3)}}{4}$ ✓
 $x = \frac{3 \pm \sqrt{15}}{2}$ ✓

(b) $2x^2 - 3x + c$ ✓
 positive definite: $a > 0$, $\Delta < 0$ ✓
 $a=2$ and $(-3)^2 - 4(2)(c) < 0$ ✓
 $9 - 8c < 0$ ✓
 $c > 1\frac{1}{8}$ ✓

(c) $2x^2 - mx + 2 = 0$ ✓
 two distinct, real roots: $\Delta > 0$ ✓
 $m^2 - 4(2)(2) > 0$ ✓
 $(m-4)(m+4) > 0$ ✓

 $m < -4$ or $m > 4$ ✓

(d) $-x^2 - 4x + 10$ ✓
 (i) $\alpha + \beta = -\frac{b}{a}$ ✓
 $= -4$ ✓

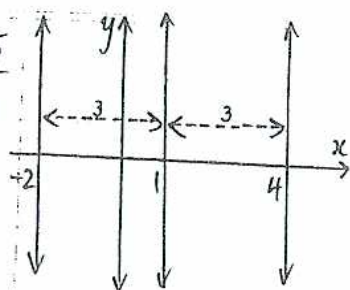
(ii) $\alpha\beta = \frac{c}{a}$ ✓
 $= -10$ ✓

(iii) $(\alpha+2)(\beta+2) = \alpha\beta + 2\alpha + 2\beta + 4$ ✓
 $= \alpha\beta + 2(\alpha+\beta) + 4$ ✓
 $= -10 + 2(-4) + 4$ ✓
 $= -14$ ✓

(iv) $\alpha^2 + \alpha\beta + \beta^2 = \alpha^2 + 2\alpha\beta + \beta^2 - \alpha\beta$ ✓
 $= (\alpha+\beta)^2 - \alpha\beta$ ✓
 $= 16 + 10$ ✓
 $= 26$ ✓

Q5

(a)



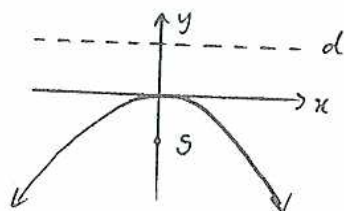
locus of P:
 $x = 4$ and $x = -2$

(b)

$$x^2 = -12y$$

$$4a = 12$$

$$a = 3$$



(i) vertex = $(0,0)$

(ii) focus = $(0,-3)$

(iii) directrix: $y = 3$

(c) (i) 15, 40, 65

(ii) $S_{41} = \frac{41}{2} (2a + 40d)$

$$= 41 (15 + 20 \times 25)$$

$$= 21115$$

(d) (i) AP: 150, $147\frac{3}{4}$, $145\frac{1}{2}$, ...

$$a = 150, d = -2\frac{1}{4}$$

$$T_n = a + (n-1)d$$

$$T_n = 150 - \frac{9}{4}(n-1)$$

$$T_n = \frac{609 - 9n}{4}$$

(ii) 1 min 50 s = 110 s

$$110 = \frac{609 - 9n}{4}$$

$$440 = 609 - 9n$$

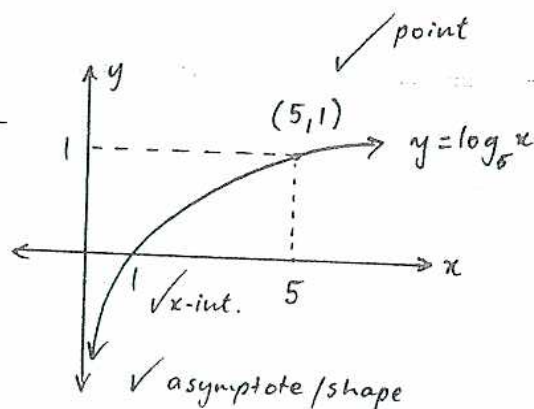
$$9n = 169$$

$$n = 18\frac{7}{9}$$

So Rory will have to run 19 races.

Q6

(a)



(b)

V(-2,1) S(-2,3)

$$(x-h)^2 = 4a(y-k)$$

$$(x+2)^2 = 8(y-1)$$

(c)

(i) $y = (x+3)^{10}$

$$y' = 10(x+3)^9$$

(ii)

$$y = \frac{x^2}{x-2}$$

$$y' = \frac{2x(x-2) - 1(x^2)}{(x-2)^2}$$

$$= \frac{2x^2 - 4x - x^2}{(x-2)^2}$$

$$= \frac{x^2 - 4x}{(x-2)^2}$$

$$= \frac{x(x-4)}{(x-2)^2}$$

(iii) (a) Let $y = 5x(x-4)^4$

$$\frac{dy}{dx} = 5(x-4)^4 + 5x \cdot 4(x-4)^3$$

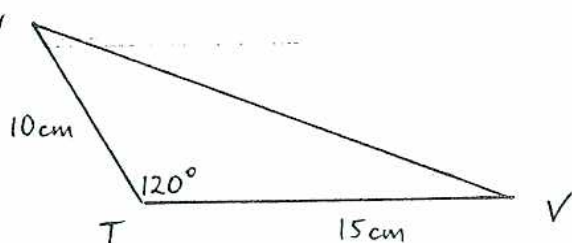
$$= 5(x-4)^3(x-4 + 4x)$$

$$= 5(5x-4)(x-4)^3$$

(b) horizontal tangent: $y' = 0$

$$x = 4 \text{ or } \frac{4}{5}$$

Q7
(a) (i)



(ii) By the cos rule,

$$UV^2 = 10^2 + 15^2 - 2 \times 10 \times 15 \cos 120^\circ \quad \checkmark$$

$$UV^2 = 325 - 300 \times -\frac{1}{2} \quad \checkmark$$

$$UV = \sqrt{475} \quad \checkmark$$

$$UV = \sqrt{25 \times 19}$$

$$UV = 5\sqrt{19}$$

(b)

$$\sum_{n=1}^{20} 3 \times 2^{n-1} = 3(2^0) + 3(2^1) + 3(2^2) + \dots + 3(2^{19}) \quad \checkmark$$

$$= 3 + 6 + 12 + \dots + 3(2^{19})$$

GP: $a=3, r=2, n=20$

$$= \frac{3(2^{20}-1)}{2-1} \quad \checkmark$$

$$= 3145725 \quad \checkmark$$

(c)

$$\frac{T_{11}}{T_4} = -\frac{512}{4}$$

$$\frac{ar^{10}}{ar^3} = -128$$

$$r^7 = -128$$

$$r = -2$$

✓ (or equations)

$$ar^3 = 4$$

$$a(-2)^3 = 4$$

$$a = -\frac{1}{2}$$

$$\begin{aligned} T_8 &= ar^7 \\ &= \left(-\frac{1}{2}\right) \times (-2)^7 \\ &= 64 \end{aligned} \quad \checkmark$$

(d)

line through the intersection: $4x-3y+2 + k(x-2y-3) = 0 \quad \checkmark$

substitute $(1, -3)$: $4+9+2 + k(1+6-3) = 0$

$$4k = -15$$

$$k = -\frac{15}{4} \quad \checkmark$$

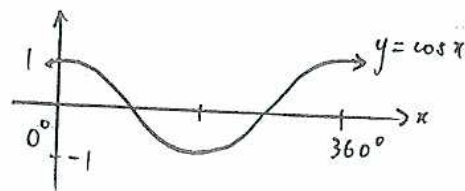
$$\text{equation: } 4x-3y+2 - \frac{15}{4}(x-2y-3) = 0$$

$$16x-12y+8 - 15x+30y+45 = 0$$

$$x + 18y + 53 = 0 \quad \checkmark$$

Q8 (a) $\cos \theta = 1$, $0^\circ \leq \theta \leq 360^\circ$

$\theta = 0^\circ$ or 360° ✓✓



(b) $2 \log_{10} x = \log_{10} (6-x)$

$\log_{10} x^2 = \log_{10} (6-x)$

$x^2 = 6-x$

$x^2 + x - 6 = 0$

$(x+3)(x-2) = 0$

$x = 2$ or -3

so

$x = 2$ is the only solution. ✓

$x > 0$ and $6-x > 0$ ✓

$0 < x < 6$

$0 < x < 6$ ✓

$0 < x < 6$

[two solutions = 2/3]

(c) (i) $y = \frac{1}{3}x^3 - \frac{4}{3}x^2$

$\frac{dy}{dx} = x^2 - \frac{8}{3}x$ ✓

when $x=3$, $\frac{dy}{dx} = 9 - \frac{8}{3} \times 3$
 $= 1$ ✓

tangent at $N(3, -3)$:

$y + 3 = 1(x - 3)$ ✓

$y = x - 6$ _____ ①

(ii) normal at $N(3, -3)$:

$y + 3 = -1(x - 3)$ ✓

$y + 3 = -x + 3$

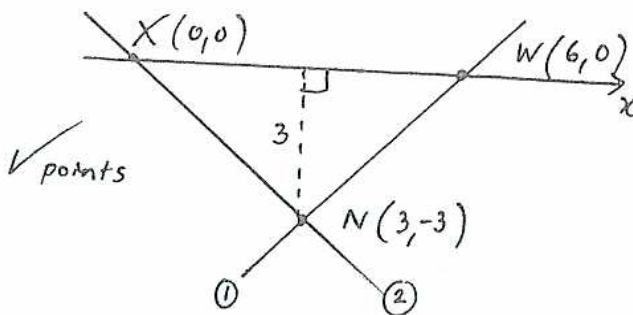
$x + y = 0$ _____ ② ✓

(iii) ① : when $y=0$, $x=6$
 $W = (6, 0)$

② : when $y=0$, $x=0$
 $X = (0, 0)$

Area of $\triangle WXN = \frac{1}{2} \times 6 \times 3$

$= 9 \text{ units}^2$



✓ area

Q9 (a)

GP:

$$r = \frac{1}{t}$$

$$S_{\infty} = \frac{1}{1-t}$$

$$\frac{a}{1 - \frac{1}{t}} = \frac{1}{1-t} \quad \checkmark$$

$$\frac{at}{t-1} = \frac{1}{1-t}$$

$$at(1-t) = t-1$$

$$a = \frac{\cancel{t-1}}{-t(\cancel{t-1})}$$

$$a = -\frac{1}{t} \quad \checkmark$$

$$GP: -\frac{1}{t}, -\frac{1}{t^2}, -\frac{1}{t^3}, \dots \quad \checkmark$$

(b)

$$y = x^2$$

$$y' = 2x$$

$$\text{when } x=c, y' = 2c \text{ and } y = c^2 \quad \checkmark$$

tangent at $x=c$:

$$y - c^2 = 2c(x - c)$$

$$y = 2cx - c^2 \quad \checkmark$$

for x -intercept $(1,0)$:

$$0 = 2c - c^2$$

$$c(2-c) = 0$$

$$c = 0 \text{ or } 2 \quad \checkmark$$

note: when $c=0$ the tangent is the x -axis, but $c \neq 0$.

solution: when $c=2$ the tangent intersects the x -axis
at $x=1$. $\checkmark$

$$(c) (i) m_{AB} = \frac{4-1}{2+1} = 1$$

$$\text{line AB: } y-1 = 1(x+1)$$

$$y-1 = x+1$$

$$x-y+2 = 0$$



(OR)

$$x-y+2 = 0$$

sub A(-1,1):

$$\text{LHS} = -1-1+2$$

$$= 0$$

$$= \text{RHS}$$

sub B(2,4):

$$\text{LHS} = 2-4+2$$

$$= 0$$

$$= \text{RHS}$$

so A, B lie on the line.

AB is the line $x-y+2 = 0$

(ii) Perpendicular distance from

$$P \text{ to AB} = \frac{|p-p^2+2|}{\sqrt{1^2+1^2}}$$

$$= \frac{|-p^2+p+2|}{\sqrt{2}}$$



$$\text{Area of } \Delta APB = \frac{1}{2} \times AB \times \text{height}$$

$$= \frac{1}{2} \times \sqrt{9+9} \times \frac{|-p^2+p+2|}{\sqrt{2}}$$

$$= \frac{1}{2} \times 3\sqrt{2} \times \frac{|-p^2+p+2|}{\sqrt{2}}$$

$$= \frac{3}{2} |-p^2+p+2| \quad \text{where } -1 < p < 2$$

The area is maximised when $-p^2+p+2$, $-1 < p < 2$ is maximised:

$$\text{i.e. when } p = -\frac{b}{2a}$$

$$p = -\frac{1}{-2}$$

$$p = \frac{1}{2}$$



Maximum area

$$\text{of } \Delta APB = \frac{3}{2} \left| -\left(\frac{1}{2}\right)^2 + \frac{1}{2} + 2 \right|$$

$$= \frac{3}{2} \times \frac{9}{4}$$

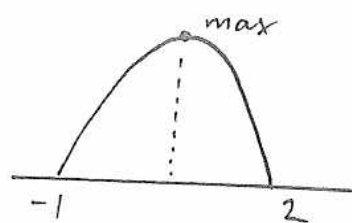
$$= 3\frac{3}{8} \text{ units}^2$$



(OR)

$$\text{Area} = \frac{3}{2} (-p^2+p+2) \quad \text{since } -1 < p < 2$$

$$= \frac{3}{2} (p+1)(2-p)$$



$$p = \frac{2-1}{2}$$

$$p = \frac{1}{2}$$

$$\text{Max. area} = \frac{3}{2} \left(\frac{1}{2} + 1 \right) \left(2 - \frac{1}{2} \right)$$

$$= 3\frac{3}{8} \text{ units}^2$$